PADHAKU
Contemporary India · Class 10 Social Science



CHAPTER 6

Manufacturing Industries

NCERT Exercise Solutions

Discuss · Manufacturing Industries	1 Q
Activity · Classification of Industries	2 Q
Discuss · Cotton Textiles	2 Q
Discuss · Sugar Industry	1 Q
Discuss · Mineral-based Industries	3 Q
Discuss · Chemical Industries	2 Q
Discuss · Cement Industry	2 Q
Discuss · Industrial Pollution and Environmental Degradation	3 Q
Exercises	6 Q
Activity · One word	7 Q
Project Work	1 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

💬 Discuss · Manufacturing Industries

? Harish saw shoes and clothes made in large industries, utensils in small industries and diyas made by artisans at home. Do you have some ideas about these industries?

💡 SAMPLE ANSWER

- **Large-scale industries:** big factories with machines and many workers, e.g. shoe, textile and sugar mills.
- **Small-scale industries:** less investment and fewer workers, e.g. units making utensils.
- **Household (cottage) industries:** run by artisans at home with simple tools and family labour, e.g. making earthen diyas.

All of them turn raw materials into more valuable finished goods.

🧪 Activity · Classification of Industries

1 List the various manufactured products you use in your daily life (transistors, electric bulbs, vegetable oil, cement, glassware, petrol, matches, scooters, medicines and so on).

💡 SAMPLE ANSWER

- Toothpaste, soap, shampoo, detergent
- Clothes, shoes, school bag, notebooks, pens, paper

- Sugar, tea, biscuits, edible oil
- Electric bulbs, fans, mobile phone, television
- Utensils, plastic bottles, glassware, matches
- Bicycle, scooter, bus, petrol
- Medicines, cement, paint

2 **Classify the following into two groups on the basis of bulk and weight of raw material and finished goods: (i) Oil (ii) Knitting needles (iii) Brassware (iv) Fuse wires (v) Watches (vi) Sewing machines (vii) Shipbuilding (viii) Electric bulbs (ix) Paint brushes (x) Automobiles**

✔ ANSWER

Heavy industries	Light industries
(i) Oil	(ii) Knitting needles
(iii) Brassware	(iv) Fuse wires
(vi) Sewing machines	(v) Watches
(vii) Shipbuilding	(viii) Electric bulbs
(x) Automobiles	(ix) Paint brushes

Heavy industries use heavy, bulky raw materials and make heavy goods; light industries use light raw materials and make light goods.

💬 Discuss · Cotton Textiles

1 **Why did Mahatma Gandhi lay emphasis on spinning yarn and weaving khadi?**

✔ ANSWER

- Khadi was a symbol of **swadeshi** and self-reliance; it was a protest against mill-made cloth from England.
- Handspun khadi gives **large-scale employment** to weavers in their homes as a cottage industry.
- It needs little money and simple tools, so poor village people could earn a living.
- It kept India's traditional skills of spinning and weaving alive.

2 **Why is it important for our country to keep the mill sector loomage lower than power loom and handloom?**

✔ ANSWER

- Handlooms and powerlooms give jobs to **many more people**, especially in villages and homes; mills use machines and fewer workers.
- Weaving is kept decentralised so that **traditional skills and designs** (cotton, silk, zari, embroidery) are used.
- It protects small weavers and cottage industries from competition with big mills.

- It helps reduce unemployment and poverty.

✏️ Discuss · Sugar Industry

? The sugar industry is seasonal in nature, so it is ideally suited to the cooperative sector. Can you explain why this is so?

✔ ANSWER

- Sugar mills work only during the **crushing season** of sugarcane, so a big private investment would stay idle for many months.
- In a cooperative, the **farmers who grow sugarcane** own the mill; they pool resources and share profits and losses.
- Farmers get an assured buyer and a fair price for their cane, and the mill gets a regular supply of raw material.
- So cooperatives are more successful, e.g. the sugar industry in Maharashtra.

💬 Discuss · Mineral-based Industries

1 Can you name some industries that would fall in the category of mineral-based industries?

✔ ANSWER

Iron and steel, aluminium smelting, cement, copper smelting, machine tools and petrochemicals.

2 **Make a list of all such goods made of steel that you can think of.**

💡 SAMPLE ANSWER

- Utensils, spoons, knives, buckets, almirahs
- Nails, screws, wires, rods for buildings, bridges
- Cars, buses, bicycles, railway coaches and tracks, ships
- Machines, tools, pipes, surgical instruments
- Weapons and defence equipment

3 **Where should the steel plants be ideally located? (Remember that the finished products also need an efficient transport network.)**

✔ ANSWER

- Near the sources of **raw materials** – iron ore, coking coal and limestone – because they are heavy and bulky, which keeps transport cost low.
- Where there is **cheap labour**, water and power supply.
- Where there is a good **transport network** (rail, road, port) to take the finished steel to markets.

- Near a large market. Example: the **Chhotanagpur plateau** region, which has the maximum concentration of iron and steel industries.

💬 Discuss · Chemical Industries

1 **Inorganic chemical industries are widely spread over the country. Why do you think it is so?**

✔ ANSWER

- Inorganic chemicals (sulphuric acid, nitric acid, soda ash, caustic soda, alkalies) are needed by many industries all over the country, e.g. fertilisers, glass, soaps, detergents, paper, paints.
- Their raw materials are widely available, and units can be both large and small.
- So they are set up near their markets everywhere.

2 **Make a list of the chemical products you are aware of.**

💡 SAMPLE ANSWER

Fertilisers, soaps and detergents, paints, dyes, plastics, synthetic fibres (nylon, polyester), synthetic rubber, medicines, adhesives, glass, insecticides and pesticides.

✏️ Discuss · Cement Industry

1 **Where would it be economically viable to set up the cement manufacturing units?**

✔ ANSWER

- Near the sources of bulky, heavy raw materials like **limestone, silica and gypsum**.
- Where **coal and electric power** are available.
- Where there is good **rail transport** and nearness to markets or ports, e.g. plants in Gujarat have access to markets in the Gulf countries.

2 **Find out where the cement plants are located in other States of India. Find their names.**

💡 SAMPLE ANSWER

Cement plants are found in limestone-rich states, for example:

- **Rajasthan** – Chittorgarh
- **Madhya Pradesh** – Katni, Satna
- **Andhra Pradesh / Telangana** – plants in the limestone areas
- **Tamil Nadu, Karnataka, Gujarat, Chhattisgarh, Jharkhand, Odisha**

Check the names of plants in your state from an atlas or the internet.

Discuss · Industrial Pollution and Environmental Degradation

1 **Are you aware of the Bhopal Gas tragedy that occurred?**

✔ ANSWER

- It happened in **Bhopal (Madhya Pradesh) in December 1984**.
- A poisonous gas (methyl isocyanate) leaked from the Union Carbide pesticide factory.
- Thousands of people died and many more suffered long-term health problems.
- It shows how hazardous toxic gas leaks from industries can be.

2 **Thermal pollution of water occurs when hot water from factories and thermal plants is drained into rivers and ponds before cooling. What would be the effect on aquatic life?**

✔ ANSWER

- Hot water reduces the **oxygen** dissolved in water, so fish and other animals can suffocate and die.
- Many aquatic plants and animals cannot bear high temperature; their breeding and growth suffer.
- The balance of the aquatic ecosystem is disturbed.

3 **How can the industrial pollution of fresh water be reduced?**

✔ ANSWER

- Minimise the use of water for processing by **reusing and recycling** it in two or more stages.
- **Harvest rainwater** to meet water needs.
- **Treat hot water and effluents** before releasing them into rivers and ponds: primary (mechanical), secondary (biological) and tertiary (biological, chemical and physical) treatment.
- Legally regulate overdrawing of ground water by industry.

📝 Exercises

1 (i) **Which one of the following industries uses bauxite as a raw material?**
(a) Aluminium Smelting (b) Cement (c) Paper (d) Steel

✔ ANSWER

(a) Aluminium Smelting – bauxite is the raw material used in aluminium smelters.

1 (ii) **Which one of the following industries manufactures telephones, computer, etc.?**

(a) Steel (b) Electronic (c) Aluminium Smelting (d) Information Technology

✔ ANSWER

(b) Electronic – the electronics industry makes telephones, computers, televisions, radars, etc.

2 (i) What is manufacturing?

✔ ANSWER

Production of goods in large quantities after processing raw materials into **more valuable products** is called manufacturing. Examples: paper from wood, sugar from sugarcane, aluminium from bauxite.

2 (ii) What are basic industries? Give an example.

✔ ANSWER

Basic or key industries supply their products as **raw materials to manufacture other goods**. Example: the **iron and steel** industry (also copper and aluminium smelting).

3 (i) How do industries pollute the environment?

✔ ANSWER

Industries help economic growth, but they cause four types of pollution: air, water, land and noise.

1. **Air pollution:** undesirable gases like sulphur dioxide and carbon monoxide, and dust and smoke from chemical and paper factories, brick kilns, refineries and smelting plants. Toxic gas leaks (e.g. Bhopal) are very hazardous.

2. **Water pollution:** paper, pulp, chemical, textile and dyeing industries, refineries, tanneries and electroplating units release dyes, detergents, acids, salts and heavy metals like lead and mercury into rivers.

3. **Thermal pollution:** hot water from factories and thermal plants is drained into rivers and ponds before cooling, harming aquatic life.

4. **Nuclear wastes** cause cancers, birth defects and miscarriages.

5. **Land pollution:** dumping of glass, harmful chemicals, effluents, packaging, salts and garbage makes soil useless; rain carries pollutants to ground water.

6. **Noise pollution:** machines, generators, saws and drills cause irritation, stress, hearing loss and higher heart rate and blood pressure.

Thus industries degrade land, water, air and human health.

3 (ii) **Discuss the steps to be taken to minimise environmental degradation by industry.**

✔ ANSWER

Economic development must go together with care for the environment. Steps:

1. **Save water:** reuse and recycle water in two or more stages; harvest rainwater.

2. **Treat effluents** before releasing them: primary (mechanical: screening, grinding, flocculation, sedimentation), secondary (biological) and tertiary (biological, chemical and physical, recycling of wastewater).

3. **Regulate by law** the overdrawing of ground water by industries.

4. **Reduce air pollution:** fit smoke stacks with electrostatic precipitators, fabric filters, scrubbers and inertial separators; use oil or gas instead of coal.

5. **Reduce noise:** fit generators with silencers, redesign machines, use noise-absorbing material and earplugs.

6. Follow examples like **NTPC**: use latest techniques, maximise ash utilisation, grow green belts, manage ash ponds and liquid waste, and monitor ecology.

In this way we can achieve sustainable development.

Activity · One word

Used to drive machinery (5) P...

✔ ANSWER

POWER

ii People who work in a factory (6) W...

✔ ANSWER
WORKER

iii Where the product is sold (6) M...

✔ ANSWER
MARKET

iv A person who sells goods (8) R...

✔ ANSWER
RETAILER

v Thing produced (7) P...

✔ ANSWER
PRODUCT

vi To make or produce (11) M...

✔ ANSWER
MANUFACTURE

vii Land, Water and Air degraded (9) P...

✔ ANSWER
POLLUTION

🛠️ Project Work

? Select one agro-based and one mineral-based industry in your area.

(i) What are the raw materials they use?

(ii) What are the other inputs in the process of manufacturing that involve transportation cost?

(iii) Are these factories following environmental norms?

💡 SAMPLE ANSWER

Model answer (use industries near you):

Question	Agro-based: Sugar mill	Mineral-based: Cement plant
(i) Raw materials	Sugarcane, water, lime	Limestone, silica, gypsum
(ii) Other inputs with transport cost	Labour, fuel (bagasse/coal), packing bags, taking sugar to markets	Coal, electricity, rail transport of raw materials and cement to markets
(iii) Environmental norms	Check effluent treatment plant and disposal of molasses/press mud	Check dust filters (electrostatic precipitators), green belt, noise control

Visit the factory with your teacher, talk to the manager and write your findings.

🧪 Activity · Word puzzle

1 Textiles, sugar, vegetable oil and plantation industries deriving raw materials from agriculture are called...

✔ ANSWER

AGRO BASED (vertical)

2 The basic raw material for sugar industry.

✔ ANSWER

SUGARCANE (horizontal)

3 This fibre is also known as the 'Golden Fibre'.

✔ ANSWER

JUTE (vertical)

4 Iron-ore, coking coal and limestone are the chief raw materials of this industry.

✔ ANSWER

IRON STEEL – iron and steel industry (horizontal)

5 A public sector steel plant located in Chhattisgarh.

✔ ANSWER

BHILAI (horizontal)

6 **Railway diesel engines are manufactured in Uttar Pradesh at this place.**

✔ ANSWER

VARANASI (horizontal)

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**